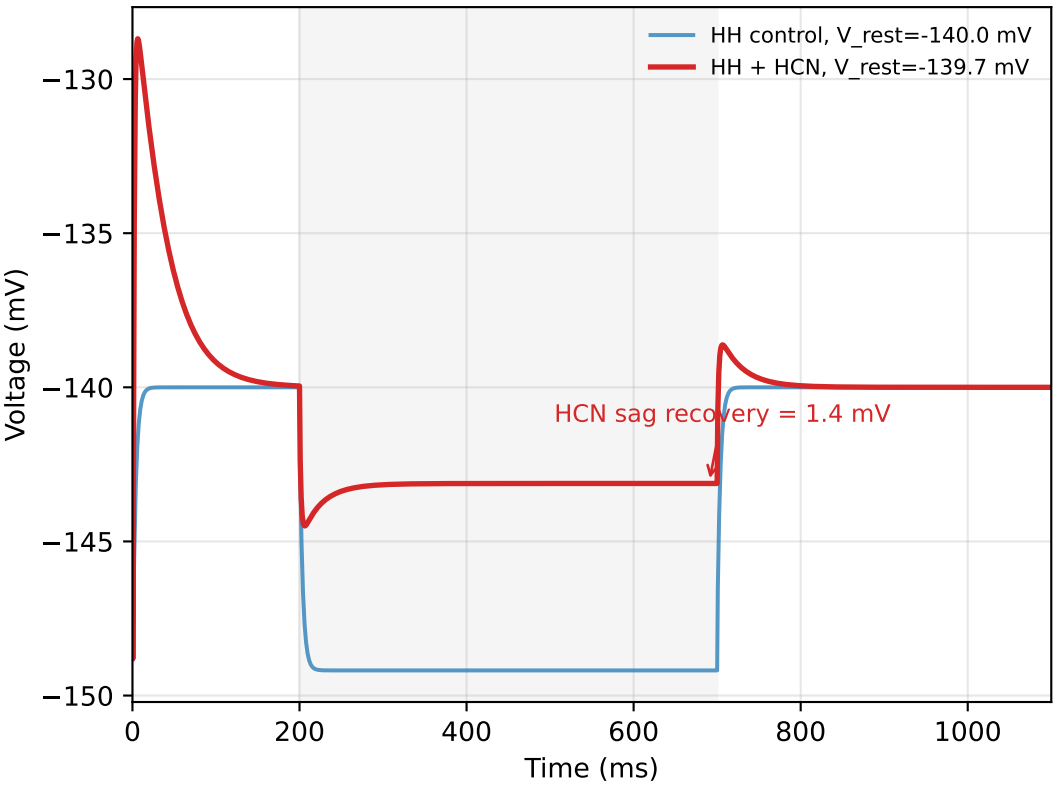
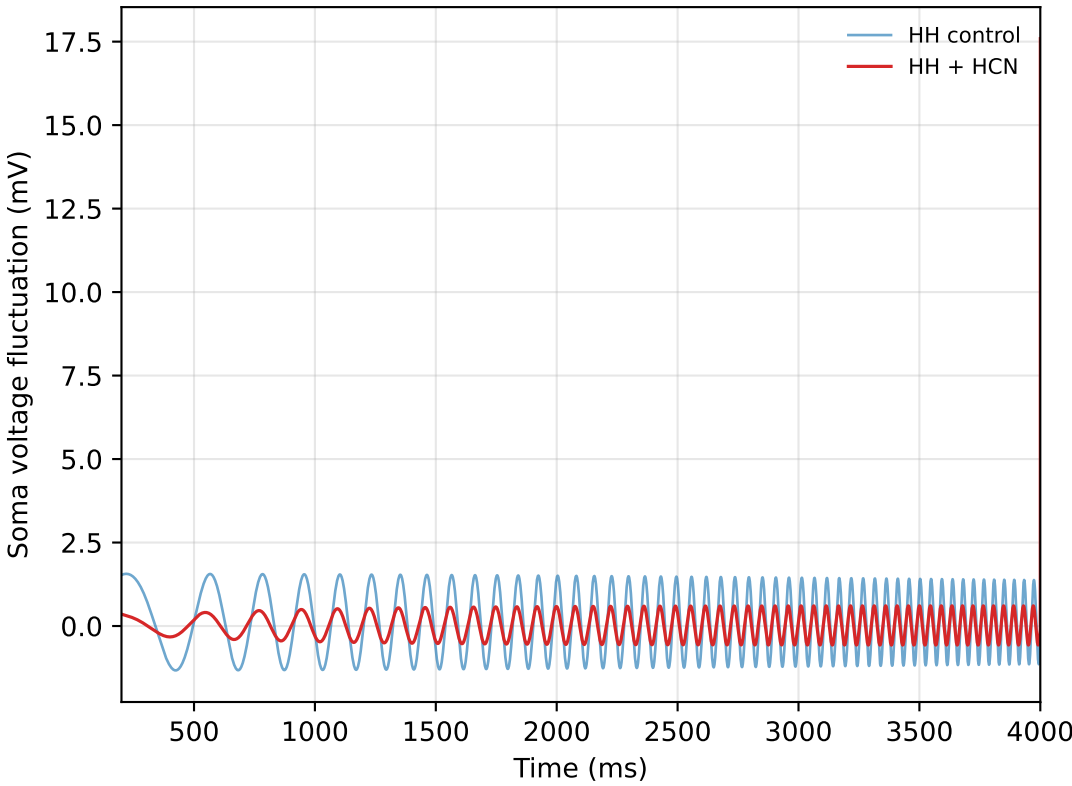


HH + HCN/Ih model: -140 mV subthreshold active resonance

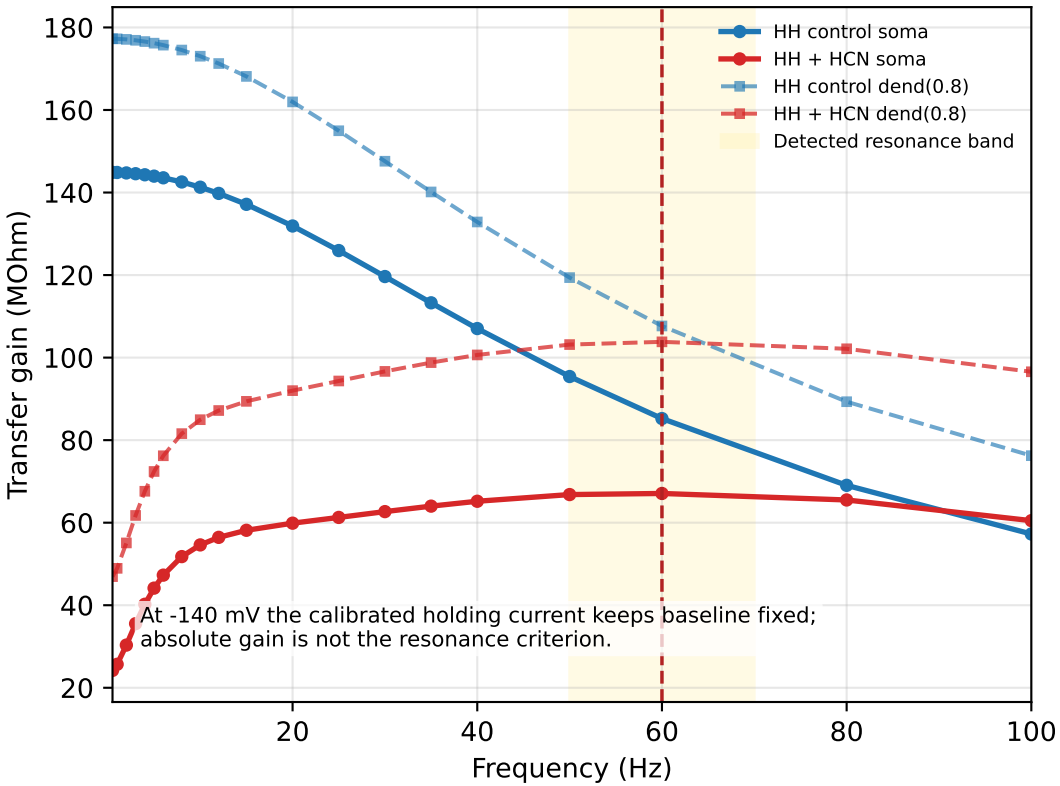
A Hyperpolarization sag with HH background



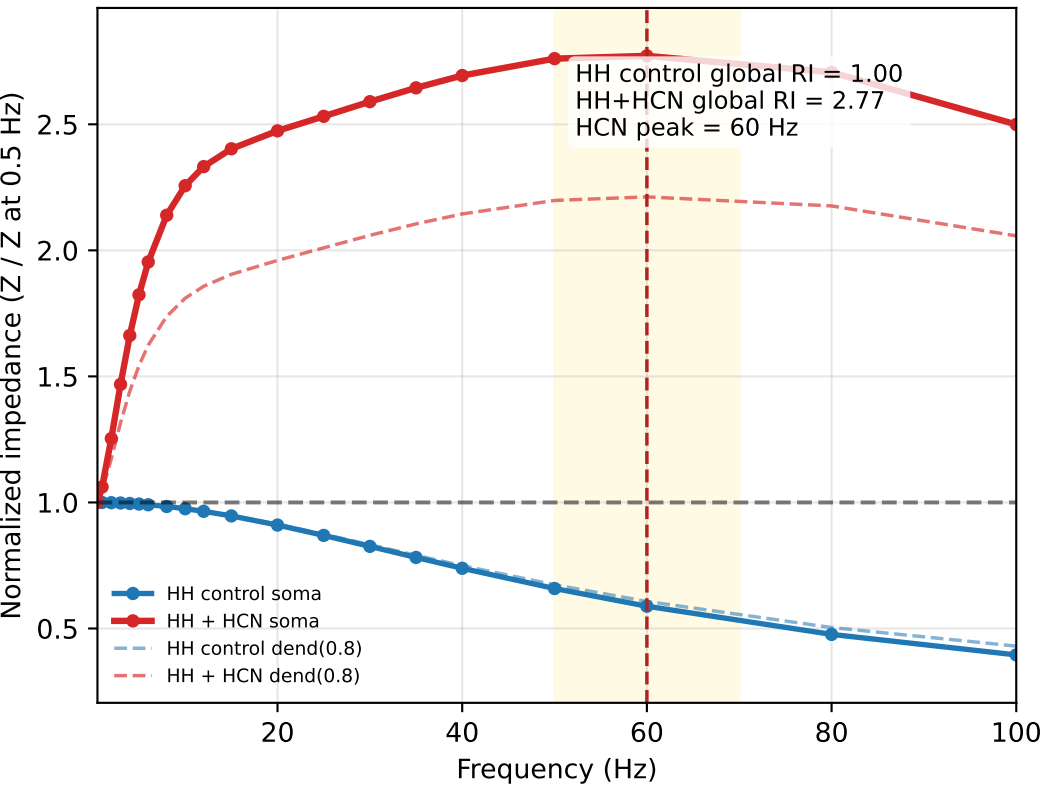
B Subthreshold ZAP trace at -140 mV, no spikes



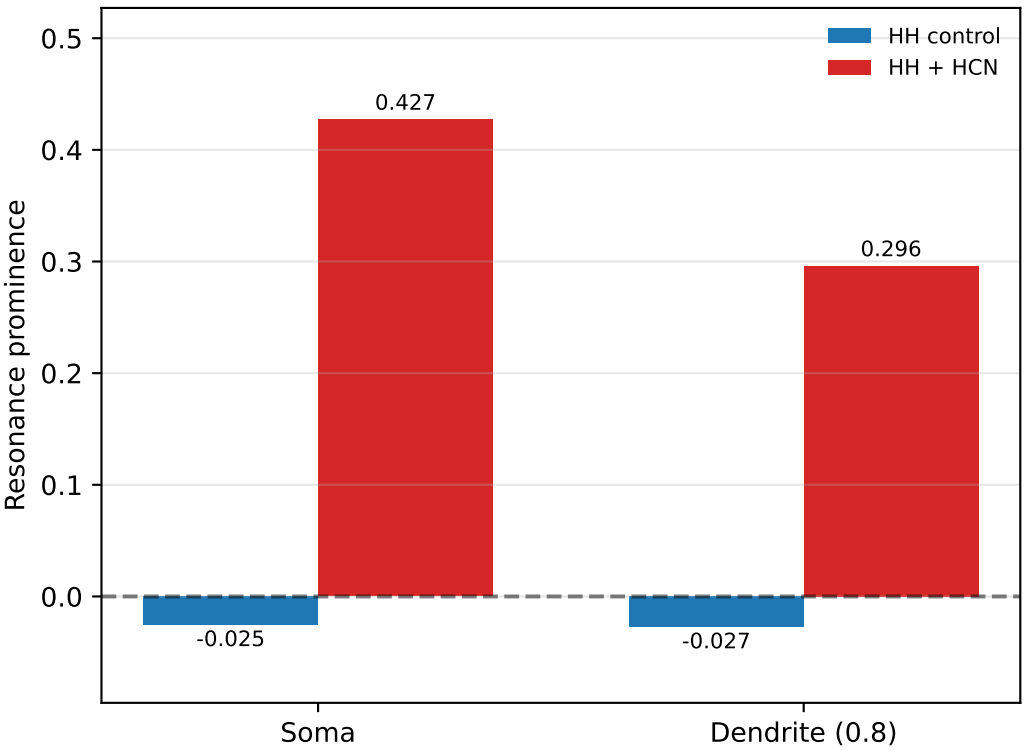
C Absolute transfer gain



D Normalized impedance / active resonance index



E Resonance prominence



Model:

- Soma: pas + hh1 Na/K +/- Ih
- Dendrite: pas +/- Ih
- Input: distal current at dend(0.8)
- Recording: soma and dend(0.8)

Operating point:

- Operating Vm: -140.0 mV
- Holding control: -0.4083 nA -> soma -140.0 mV, dend -124.2 mV
- Holding HCN: -0.8865 nA -> soma -140.0 mV, dend -108.0 mV
- Holding: calibrated current clamp, not voltage clamp
- Fixed g_h: 0.0008 S/cm2
- Soma Ih density: 0.25 x g_h
- AC amplitude: 0.0100 nA
- Frequency range: 0.5-100 Hz
- Resonance band: 50.0-70.0 Hz
- HH effects are expected to be weak at this hyperpolarized Vm.

Validation:

- HH spike validation: True, current=0.08 nA
- ZAP/sweep for resonance: no spikes = True

Results:

- HCN sag recovery: 1.38 mV
- HH control global RI: 1.000
- HH+HCN global RI: 2.772
- HH+HCN global peak: 60.0 Hz
- Theta-band RI: 2.256
- Prominence soma: control=-0.025, HCN=0.427
- Prominence dend: control=-0.027, HCN=0.296

Conclusion:

HH+HCN global RI > HH control global RI under -140 mV subthreshold conditions. This reflects normalized frequency tuning, not absolute voltage amplification.